Class 10 Mathematics — Exercise 2.4

Question 1

Find the sum and product of the roots of the quadratic equation.

(i) $x^2 - 5x + 2 = 0$

$a = 1, \quad b = -5, \quad c = 2$

Let S_1 and S_2 be the roots.

Sum of roots:

$$\text{Sum of roots} = S_1 + S_2 = -\frac{b}{a} = -\frac{(-5)}{1}$$

$$S = 5$$

Product of roots:

$$\text{Product of roots} = S_1 S_2 = \frac{c}{a} = \frac{2}{1}$$

$$P = 2$$

Result: $\boxed{S = 5, \quad P = 2}$

(ii) $-4x^2 - 6x - 2 = 0$

$a = -4, \quad b = -6, \quad c = -2$

Let S_1 and S_2 be the roots.

Sum of roots:

$$\text{Sum of roots} = S_1 + S_2 = -\frac{b}{a} = -\frac{(-6)}{-4}$$

$$S = -\frac{3}{2}$$

Product of roots:

$$\text{Product of roots} = S_1 S_2 = \frac{c}{a} = \frac{-2}{-4}$$

$$P = \frac{1}{2}$$

Result: $\boxed{S = -\frac{3}{2}, \quad P = \frac{1}{2}}$

(iii) $5x^2 - 2x + 2 = 0$

$a = 5, \quad b = -2, \quad c = 2$

Let S_1 and S_2 be the roots.

Sum of roots:

$$\text{Sum of roots} = S_1 + S_2 = -\frac{b}{a} = -\frac{(-2)}{5}$$

$$S = \frac{2}{5}$$

Product of roots:

$$\text{Product of roots} = S_1 S_2 = \frac{c}{a} = \frac{2}{5}$$

$$P = \frac{2}{5}$$

Result: $\boxed{S = \frac{2}{5}, \quad P = \frac{2}{5}}$

(iv) $-4x^2 - 8x - 9 = 0$

$a = -4, \quad b = -8, \quad c = -9$

Let S_1 and S_2 be the roots.

Sum of roots:

$$\text{Sum of roots} = S_1 + S_2 = -\frac{b}{a} = -\frac{(-8)}{-4}$$

$$S = -2$$

Product of roots:

$$\text{Product of roots} = S_1 S_2 = \frac{c}{a} = \frac{-9}{-4}$$

$$P = \frac{9}{4}$$

Result: $\boxed{S = -2, \quad P = \frac{9}{4}}$

(v) $16y^2 - 17y - 12 = 0$

$a = 16, \quad b = -17, \quad c = -12$

Let S_1 and S_2 be the roots.

Sum of roots:

$$\text{Sum of roots} = S_1 + S_2 = -\frac{b}{a} = -\frac{(-17)}{16}$$

$$S = \frac{17}{16}$$

Product of roots:

$$\text{Product of roots} = S_1 S_2 = \frac{c}{a} = \frac{-12}{16}$$

$$P = -\frac{3}{4}$$

Result: $\boxed{S = \frac{17}{16}, \quad P = -\frac{3}{4}}$

(vi) $0.3x^2 - 7.7x + 1.8 = 0$

$a = 0.3, \quad b = -7.7, \quad c = 1.8$

Let S_1 and S_2 be the roots.

Sum of roots:

$$\text{Sum of roots} = S_1 + S_2 = -\frac{b}{a} = -\frac{(-7.7)}{0.3}$$

$$S = 25.67$$

Product of roots:

$$\text{Product of roots} = S_1 S_2 = \frac{c}{a} = \frac{1.8}{0.3}$$

$$P = 6$$

Result: $\boxed{S = 25.67, \quad P = 6}$

Question 2

Form a quadratic equation with roots:

In each part we use

$$x^2 - Sx + P = 0 \qquad .....(1)$$

where S = sum of roots and P = product of roots.

(i) $1, \ -\frac{8}{3}$ **(Given roots)**

S = Sum of roots:

$$S = (1) + \left(-\frac{8}{3}\right) = \frac{1}{1} - \frac{8}{3} = \frac{3-8}{3}$$

$$S = -\frac{5}{3}$$

P = Product of roots:

$$P = (1)\left(-\frac{8}{3}\right)$$

$$P = -\frac{8}{3}$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - \left(-\frac{5}{3}\right)x + \left(-\frac{8}{3}\right) = 0$$

$$x^2 + \frac{5}{3}x - \frac{8}{3} = 0$$

Multiply by 3 on both sides:

$$3x^2 + 3 \times \frac{5}{3}x - 3 \times \frac{8}{3} = 0 \times 3$$

$$\boxed{3x^2 + 5x - 8 = 0}$$

(ii) $\sqrt{3},\ 2\sqrt{3}$ (Given roots)

S = Sum of roots:

$$S = \sqrt{3} + 2\sqrt{3}$$

$$S = 3\sqrt{3}$$

P = Product of roots:

$$P = (\sqrt{3})(2\sqrt{3}) = 2(\sqrt{3})^2$$

$$P = 6$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$\boxed{x^2 - 3\sqrt{3}\,x + 6 = 0}$$

(iii) $2+\sqrt{3},\ 2-\sqrt{3}$ (Given roots)

S = Sum of roots:

$$S = 2+\sqrt{3}+2-\sqrt{3}$$

$$S = 4$$

P = Product of roots:

Using $(a+b)(a-b) = a^2 - b^2$:

$$P = (2+\sqrt{3})(2-\sqrt{3}) = (2)^2 - (\sqrt{3})^2 = 4-3$$

$$P = 1$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$\boxed{x^2 - 4x + 1 = 0}$$

(iv) $5i,\ -5i$ (Given roots)

S = Sum of roots:

$$S = (5i) + (-5i) = 5i - 5i$$

$$S = 0$$

P = Product of roots:

$$P = (5i)(-5i) = -25i^2 = -25(-1)$$

$$P = 25$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - 0x + 25 = 0$$

$$\boxed{x^2 + 25 = 0}$$

(v) $7+2i,\ 7-2i$ (Given roots)

S = Sum of roots:

$$S = 7 + 2i + 7 - 2i$$

$$S = 14$$

P = Product of roots:

$$P = (7+2i)(7-2i) = (7)^2 - (2i)^2 = 49 - 4i^2 = 49 - 4(-1) = 49 + 4$$

$$P = 53$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$\boxed{x^2 - 14x + 53 = 0}$$

Question 3

If S_1, S_2 are the roots of $3x^2 - 2x + 4 = 0$, find the value of:

$$3x^2 - 2x + 4 = 0$$

$a = 3, \quad b = -2, \quad c = 4$

$$S_1 + S_2 = -\frac{b}{a} = -\frac{(-2)}{3} = \frac{2}{3}$$

$$S_1 S_2 = \frac{c}{a} = \frac{4}{3}$$

Formula used:

$$(S_1 + S_2)^2 = S_1^2 + S_2^2 + 2S_1S_2$$

$$\boxed{(S_1 + S_2)^2 - 2S_1S_2 = S_1^2 + S_2^2}$$

(i) $\dfrac{1}{S_1^2} + \dfrac{1}{S_2^2}$

$$= \frac{S_2^2 + S_1^2}{S_1^2 S_2^2}$$

$$= \frac{S_1^2 + S_2^2 + 2S_1S_2 - 2S_1S_2}{(S_1S_2)^2}$$

$$= \frac{(S_1 + S_2)^2 - 2S_1S_2}{(S_1S_2)^2}$$

$$= \frac{\left(\frac{2}{3}\right)^2 - 2\left(\frac{4}{3}\right)}{\left(\frac{4}{3}\right)^2}$$

$$= \frac{\frac{4}{9} - \frac{8}{3}}{\frac{16}{9}}$$

$$= \frac{4 - 24}{9} \div \frac{16}{9}$$

$$= -\frac{20}{9} \times \frac{9}{16}$$

$$= \boxed{-\frac{5}{4}}$$

(ii) $S_1^2 + S_2^2$

$$= S_1^2 + S_2^2 + 2S_1S_2 - 2S_1S_2$$

$$= (S_1 + S_2)^2 - 2S_1S_2$$

$$= \left(\frac{2}{3}\right)^2 - 2\left(\frac{4}{3}\right)$$

$$= \frac{4}{9} - \frac{8}{3}$$

$$= \frac{4 - 24}{9}$$

$$= \boxed{-\frac{20}{9}}$$

(iii) $2S_1 + 2S_2 + 4$

$$= 2(S_1 + S_2) + 4$$

$$= 2\left(\frac{2}{3}\right) + 4$$

$$= \frac{4}{3} + \frac{4}{1}$$

$$= \frac{4 + 12}{3}$$

$$= \boxed{\frac{16}{3}}$$

(iv) $\dfrac{1}{S_1}+\dfrac{1}{S_2}$

$$= \frac{S_2+S_1}{S_1S_2}$$

$$= \frac{\frac{2}{3}}{\frac{4}{3}}$$

$$= \frac{2}{3} \div \frac{4}{3}$$

$$= \frac{2}{3} \times \frac{3}{4}$$

$$= \boxed{\frac{1}{2}}$$

(v) $\dfrac{S_1}{S_2}+\dfrac{S_2}{S_1}$

$$= \frac{S_1^2+S_2^2}{S_1S_2}$$

$$= \frac{S_1^2+S_2^2+2S_1S_2-2S_1S_2}{S_1S_2}$$

$$= \frac{(S_1+S_2)^2-2S_1S_2}{S_1S_2}$$

$$= \frac{\left(\frac{2}{3}\right)^2-2\left(\frac{4}{3}\right)}{\frac{4}{3}}$$

$$= \left(\frac{4}{9}-\frac{8}{3}\right) \div \frac{4}{3}$$

$$= \frac{4-24}{9} \times \frac{3}{4}$$

$$= -\frac{20}{9} \times \frac{3}{4}$$

$$= \boxed{-\frac{5}{3}}$$

(vi) $S_1S_2^2+S_1^2S_2$

$$= S_1S_2(S_2+S_1)$$

$$= \frac{4}{3}\left(\frac{2}{3}\right)$$

$$= \boxed{\frac{8}{9}}$$

(vii) $S_1^3S_2 + S_1S_2^3$

$$= S_1S_2(S_1^2 + S_2^2)$$

$$= S_1S_2\left[S_1^2 + S_2^2 + 2S_1S_2 - 2S_1S_2\right]$$

$$= S_1S_2\left[(S_1 + S_2)^2 - 2S_1S_2\right]$$

$$= \frac{4}{3}\left[\left(\frac{2}{3}\right)^2 - 2\left(\frac{4}{3}\right)\right]$$

$$= \frac{4}{3}\left[\frac{4}{9} - \frac{8}{3}\right]$$

$$= \frac{4}{3}\left[\frac{4-24}{9}\right]$$

$$= \frac{4}{3}\left[-\frac{20}{9}\right]$$

$$= \boxed{-\frac{80}{27}}$$

(viii) $(S_1 - 3)(S_2 - 3)$

$$= S_1(S_2 - 3) - 3(S_2 - 3)$$

$$= S_1S_2 - 3S_1 - 3S_2 + 9$$

$$= S_1S_2 - 3(S_1 + S_2) + 9$$

$$= \frac{4}{3} - 3\left(\frac{2}{3}\right) + 9$$

$$= \frac{4}{3} - 2 + 9$$

$$= \frac{4}{3} + \frac{7}{1}$$

$$= \frac{4 + 21}{3}$$

$$= \boxed{\frac{25}{3}}$$

Question 4

If S_1, S_2 are the roots of $7x^2 + 10x + 7 = 0$, form the equations whose roots are:

$$7x^2 + 10x + 7 = 0 \qquad \text{(Given equation)}$$

$a = 7, \quad b = 10, \quad c = 7$

$S_1,\ S_2$ (Given roots)

$$S_1 + S_2 = -\frac{b}{a} = -\frac{10}{7}$$

$$S_1 S_2 = \frac{c}{a} = \frac{7}{7} = 1$$

Required equation:

$$x^2 - Sx + P = 0 \qquad .....(1)$$

(i) $S_1^2,\ S_2^2$

S:

$$S = S_1^2 + S_2^2$$

$$= S_1^2 + S_2^2 + 2S_1S_2 - 2S_1S_2$$

$$= (S_1 + S_2)^2 - 2S_1S_2$$

$$= \left(-\frac{10}{7}\right)^2 - 2(1)$$

$$= \frac{100}{49} - \frac{2}{1}$$

$$= \frac{100 - 98}{49}$$

$$S = \frac{2}{49}$$

P:

$$P = (S_1^2)(S_2^2) = (S_1S_2)^2 = (1)^2$$

$$P = 1$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - \frac{2}{49}x + 1 = 0$$

Multiply by 49 on both sides:

$$49x^2 - 49 \times \frac{2}{49}x + 49 \times 1 = 0 \times 49$$

$$\boxed{49x^2 - 2x + 49 = 0}$$

(ii) $\frac{1}{S_1},\ \frac{1}{S_2}$

S:

$$S = \frac{1}{S_1} + \frac{1}{S_2} = \frac{S_2 + S_1}{S_1 S_2} = \frac{-\frac{10}{7}}{1}$$

$$S = -\frac{10}{7}$$

P:

$$P = \left(\frac{1}{S_1}\right)\left(\frac{1}{S_2}\right) = \frac{1}{S_1 S_2} = \frac{1}{1}$$

$$P = 1$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - \left(-\frac{10}{7}\right)x + 1 = 0$$

$$x^2 + \frac{10}{7}x + 1 = 0$$

Multiply by 7 on both sides:

$$7x^2 + 7 \times \frac{10}{7}x + 7 \times 1 = 0 \times 7$$

$$\boxed{7x^2 + 10x + 7 = 0}$$

(iii) $S_1^3 S_2,\ S_1 S_2^3$

S:

$$S = S_1^3 S_2 + S_1 S_2^3$$

$$= S_1 S_2 (S_1^2 + S_2^2)$$

$$= S_1S_2(S_1^2 + S_2^2 + 2S_1S_2 - 2S_1S_2)$$

$$= S_1S_2\left((S_1 + S_2)^2 - 2S_1S_2\right)$$

$$= 1\left[\left(-\frac{10}{7}\right)^2 - 2(1)\right]$$

$$= \left[\frac{100}{49} - \frac{2}{1}\right]$$

$$= \frac{100 - 98}{49}$$

$$S = \frac{2}{49}$$

P:

$$P = (S_1^3S_2)(S_1S_2^3) = S_1^4S_2^4 = (S_1S_2)^4 = (1)^4$$

$$P = 1$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - \frac{2}{49}x + 1 = 0$$

Multiply by 49 on both sides:

$$49x^2 - 49 \times \frac{2}{49}x + 49 \times 1 = 49 \times 0$$

$$\boxed{49x^2 - 2x + 49 = 0}$$

(iv) $S_1 - \frac{1}{S_1},\ S_2 - \frac{1}{S_2}$

S:

$$S = S_1 - \frac{1}{S_1} + S_2 - \frac{1}{S_2}$$

$$= S_1 + S_2 - \left(\frac{1}{S_1} + \frac{1}{S_2}\right)$$

$$= S_1 + S_2 - \left(\frac{S_2 + S_1}{S_1S_2}\right)$$

$$= -\frac{10}{7} - \left(\frac{-\frac{10}{7}}{1}\right)$$

$$= -\frac{10}{7} + \frac{10}{7}$$

$$S = 0$$

P:

$$P = \left(S_1 - \frac{1}{S_1}\right)\left(S_2 - \frac{1}{S_2}\right)$$

$$= S_1\left(S_2 - \frac{1}{S_2}\right) - \frac{1}{S_1}\left(S_2 - \frac{1}{S_2}\right)$$

$$= S_1S_2 - \frac{S_1}{S_2} - \frac{S_2}{S_1} + \frac{1}{S_1S_2}$$

$$= S_1S_2 + \frac{1}{S_1S_2} - \left[\frac{S_1}{S_2} + \frac{S_2}{S_1}\right]$$

$$= S_1S_2 + \frac{1}{S_1S_2} - \left[\frac{S_1^2 + S_2^2}{S_1S_2}\right]$$

$$= S_1S_2 + \frac{1}{S_1S_2} - \left[\frac{(S_1 + S_2)^2 - 2S_1S_2}{S_1S_2}\right]$$

$$= 1 + \frac{1}{1} - \left[\frac{\left(-\frac{10}{7}\right)^2 - 2(1)}{1}\right]$$

$$= 1 + 1 - \left[\frac{100}{49} - \frac{2}{1}\right]$$

$$= 2 - \left[\frac{100 - 98}{49}\right]$$

$$= \frac{2}{1} - \frac{2}{49}$$

$$= \frac{98 - 2}{49}$$

$$P = \frac{96}{49}$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - 0x + \frac{96}{49} = 0$$

$$x^2 + \frac{96}{49} = 0$$

Multiply by 49 on both sides:

$$49x^2 + 49 \times \frac{96}{49} = 49 \times 0$$

$$\boxed{49x^2 + 96 = 0}$$

(v) $2S_1 + 1,\ 2S_2 + 1$

S:

$$S = 2S_1 + 1 + 2S_2 + 1$$

$$= 2(S_1 + S_2) + 2$$

$$= 2\left(-\frac{10}{7}\right) + 2$$

$$= -\frac{20}{7} + \frac{2}{1}$$

$$= \frac{-20 + 14}{7}$$

$$S = -\frac{6}{7}$$

P:

$$P = (2S_1 + 1)(2S_2 + 1)$$

$$= 2S_1(2S_2 + 1) + 1(2S_2 + 1)$$

$$= 4S_1S_2 + 2S_1 + 2S_2 + 1$$

$$= 4S_1S_2 + 2(S_1 + S_2) + 1$$

$$= 4(1) + 2\left(-\frac{10}{7}\right) + 1$$

$$= 4 - \frac{20}{7} + 1$$

$$= \frac{5}{1} - \frac{20}{7}$$

$$= \frac{35 - 20}{7}$$

$$P = \frac{15}{7}$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - \left(-\frac{6}{7}\right) x + \frac{15}{7} = 0$$

$$x^2 + \frac{6}{7}x + \frac{15}{7} = 0$$

Multiply by 7 on both sides:

$$7x^2 + 7 \times \frac{6}{7}x + 7 \times \frac{15}{7} = 0 \times 7$$

$$\boxed{7x^2 + 6x + 15 = 0}$$

(vi) $\frac{S_1}{S_2}, \frac{S_2}{S_1}$

S:

$$S = \frac{S_1}{S_2} + \frac{S_2}{S_1}$$

$$= \frac{S_1^2 + S_2^2}{S_1 S_2}$$

$$= \frac{(S_1 + S_2)^2 - 2S_1 S_2}{S_1 S_2}$$

$$= \frac{\left(-\frac{10}{7}\right)^2 - 2(1)}{1}$$

$$= \frac{100}{49} - \frac{2}{1}$$

$$= \frac{100 - 98}{49}$$

$$S = \frac{2}{49}$$

P:

$$P = \left(\frac{S_1}{S_2}\right)\left(\frac{S_2}{S_1}\right) = \frac{1}{1}$$

$$P = 1$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - \frac{2}{49}x + 1 = 0$$

Multiply by 49 on both sides:

$$49x^2 - 49 \times \frac{2}{49}x + 49 \times 1 = 0 \times 49$$

$$\boxed{49x^2 - 2x + 49 = 0}$$

(vii) $S_1 + S_2,\ \frac{1}{S_1} + \frac{1}{S_2}$

S:

$$S = S_1 + S_2 + \frac{1}{S_1} + \frac{1}{S_2}$$

$$= S_1 + S_2 + \left(\frac{S_2 + S_1}{S_1 S_2}\right)$$

$$= -\frac{10}{7} + \left(\frac{-\frac{10}{7}}{1}\right)$$

$$= -\frac{10}{7} - \frac{10}{7}$$

$$= \frac{-10 - 10}{7}$$

$$S = -\frac{20}{7}$$

P:

$$P = (S_1 + S_2)\left(\frac{1}{S_1} + \frac{1}{S_2}\right)$$

$$= (S_1 + S_2)\left(\frac{S_2 + S_1}{S_1 S_2}\right)$$

$$= -\frac{10}{7}\left(\frac{-\frac{10}{7}}{1}\right)$$

$$= -\frac{10}{7}\left(-\frac{10}{7}\right)$$

$$P = \frac{100}{49}$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - \left(-\frac{20}{7}\right)x + \frac{100}{49} = 0$$

$$x^2 + \frac{20}{7}x + \frac{100}{49} = 0$$

Multiply by 49 on both sides:

$$49x^2 + 49 \times \frac{20}{7}x + 49 \times \frac{100}{49} = 49 \times 0$$

$$\boxed{49x^2 + 140x + 100 = 0}$$

(viii) $S_1^2 + 1,\ S_2^2 + 1$

S:

$$S = S_1^2 + 1 + S_2^2 + 1$$

$$= S_1^2 + S_2^2 + 2$$

$$= (S_1 + S_2)^2 - 2S_1S_2 + 2$$

$$= \left(-\frac{10}{7}\right)^2 - 2(1) + 2$$

$$= \frac{100}{49} - 2 + 2$$

$$S = \frac{100}{49}$$

P:

$$P = (S_1^2 + 1)(S_2^2 + 1)$$

$$= S_1^2(S_2^2 + 1) + 1(S_2^2 + 1)$$

$$= S_1^2 S_2^2 + S_1^2 + S_2^2 + 1$$

$$= (S_1 S_2)^2 + (S_1 + S_2)^2 - 2S_1 S_2 + 1$$

$$= (1)^2 + \left(-\frac{10}{7}\right)^2 - 2(1) + 1$$

$$= 1 + \frac{100}{49} - 2 + 1$$

$$= \frac{100}{49} - 2 + 2$$

$$P = \frac{100}{49}$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - \frac{100}{49}x + \frac{100}{49} = 0$$

Multiply by 49 on both sides:

$$49x^2 - 49 \times \frac{100}{49}x + 49 \times \frac{100}{49} = 0 \times 49$$

$$\boxed{49x^2 - 100x + 100 = 0}$$

(ix) $S_1^2 + S_2,\ S_1 + S_2^2$
S:

$$S = S_1^2 + S_2 + S_1 + S_2^2$$

$$= S_1^2 + S_2^2 + S_1 + S_2$$

$$= (S_1 + S_2)^2 - 2S_1 S_2 + S_1 + S_2$$

$$= \left(-\frac{10}{7}\right)^2 - 2(1) + \left(-\frac{10}{7}\right)$$

$$= \frac{100}{49} - \frac{2}{1} - \frac{10}{7}$$

$$= \frac{100 - 98 - 70}{49}$$

$$S = -\frac{68}{49}$$

P:

$$P = (S_1^2 + S_2)(S_1 + S_2^2)$$

$$= S_1^2(S_1 + S_2^2) + S_2(S_1 + S_2^2)$$

$$= S_1^3 + S_1^2 S_2^2 + S_1 S_2 + S_2^3$$

$$= S_1^3 + S_2^3 + (S_1 S_2)^2 + S_1 S_2$$

$$\because (a^3 + b^3) = (a + b)(a^2 + b^2 - ab)$$

$$= (S_1 + S_2)\left[S_1^2 + S_2^2 - S_1 S_2\right] + (S_1 S_2)^2 + S_1 S_2$$

$$= (S_1 + S_2)\left[(S_1 + S_2)^2 - 2S_1 S_2 - S_1 S_2\right] + (S_1 S_2)^2 + S_1 S_2$$

$$= -\frac{10}{7}\left[\left(-\frac{10}{7}\right)^2 - 2(1) - 1\right] + (1)^2 + 1$$

$$= -\frac{10}{7}\left[\frac{100}{49} - 2 - 1\right] + 1 + 1$$

$$= -\frac{10}{7}\left[\frac{100}{49} - \frac{3}{1}\right] + 2$$

$$= -\frac{10}{7}\left[\frac{100 - 147}{49}\right] + 2$$

$$= -\frac{10}{7}\left[-\frac{47}{49}\right] + 2$$

$$= \frac{470}{343} + \frac{2}{1}$$

$$= \frac{470 + 686}{343}$$

$$P = \frac{1156}{343}$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - \left(-\frac{68}{49}\right)x + \frac{1156}{343} = 0$$

$$x^2 + \frac{68}{49}x + \frac{1156}{343} = 0$$

Multiply by 343 on both sides (LCM of 49 and 343 is 343):

$$343x^2 + 343 \times \frac{68}{49}x + 343 \times \frac{1156}{343} = 0$$

$$\boxed{343x^2 + 476x + 1156 = 0}$$

Question 5

If S_1, S_2 are the roots of $x^2 + 6x + 3 = 0$, form the equation whose roots are $(S_1 + S_2)^2,\ (S_1 - S_2)^2$.

Given equation:

$$x^2 + 6x + 3 = 0$$

$a = 1, \quad b = 6, \quad c = 3$

$$S_1 + S_2 = -\frac{b}{a} = -\frac{6}{1} = -6$$

$$S_1 S_2 = \frac{c}{a} = \frac{3}{1} = 3$$

Required equation:

$$x^2 - Sx + P = 0 \qquad .....(1)$$

$(S_1 + S_2)^2,\ (S_1 - S_2)^2$ (Given roots)

S:

$$S = (S_1 + S_2)^2 + (S_1 - S_2)^2$$

$$= (S_1 + S_2)^2 + [S_1^2 + S_2^2 - 2S_1S_2]$$

$$= (S_1 + S_2)^2 + [(S_1 + S_2)^2 - 2S_1S_2 - 2S_1S_2]$$

$$= (S_1 + S_2)^2 + [(S_1 + S_2)^2 - 4S_1S_2]$$

$$= (-6)^2 + [(-6)^2 - 4(3)]$$

$$= 36 + 36 - 12$$

$$S = 60$$

P:

$$P = (S_1 + S_2)^2 (S_1 - S_2)^2$$

$$= (S_1 + S_2)^2 \left[S_1^2 + S_2^2 - 2S_1S_2\right]$$

$$= (S_1 + S_2)^2 \left[(S_1 + S_2)^2 - 2S_1S_2 - 2S_1S_2\right]$$

$$= (S_1 + S_2)^2 \left[(S_1 + S_2)^2 - 4S_1S_2\right]$$

$$= (-6)^2 \left[(-6)^2 - 4(3)\right]$$

$$= 36 \left[36 - 12\right]$$

$$= 36 \left[24\right]$$

$$P = 864$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$\boxed{x^2 - 60x + 864 = 0}$$

Question 6

If S_1, S_2 are the roots of $2x^2 + 6x - 3 = 0$, form the equation whose roots are $S_1 - \frac{3}{S_2^2},\ S_2 - \frac{3}{S_1^2}$.

$$2x^2 + 6x - 3 = 0$$

$a = 2, \quad b = 6, \quad c = -3$

$$S_1 + S_2 = -\frac{b}{a} = -\frac{6}{2} = -3$$

$$S_1 S_2 = \frac{c}{a} = -\frac{3}{2}$$

$$x^2 - Sx + P = 0 \qquad .....(1)$$

$S_1 - \frac{3}{S_2^2},\ S_2 - \frac{3}{S_1^2}$ (Given roots)

S:

$$S = S_1 - \frac{3}{S_2^2} + S_2 - \frac{3}{S_1^2}$$

$$= S_1 + S_2 - 3\left[\frac{1}{S_2^2} + \frac{1}{S_1^2}\right]$$

$$= S_1 + S_2 - 3\left[\frac{S_1^2 + S_2^2}{S_1^2 S_2^2}\right]$$

$$= S_1 + S_2 - 3\left[\frac{(S_1 + S_2)^2 - 2S_1 S_2}{(S_1 S_2)^2}\right]$$

$$= -3 - 3\left[\frac{(-3)^2 - 2\left(-\frac{3}{2}\right)}{\left(-\frac{3}{2}\right)^2}\right]$$

$$= -3 - 3\left[9 + 3 \div \left(-\frac{3}{2}\right)^2\right]$$

$$= -3 - 3\left[12 \div \frac{9}{4}\right]$$

$$= -3 - 3\left[12 \times \frac{4}{9}\right]$$

$$= -3 - 3\left[\frac{16}{3}\right]$$

$$= -3 - 16$$

$$S = -19$$

P:

$$P = \left(S_1 - \frac{3}{S_2^2}\right)\left(S_2 - \frac{3}{S_1^2}\right)$$

$$= S_1\left(S_2 - \frac{3}{S_1^2}\right) - \frac{3}{S_2^2}\left(S_2 - \frac{3}{S_1^2}\right)$$

$$= S_1S_2 - \frac{3S_1}{S_1^2} - \frac{3S_2}{S_2^2} + \frac{9}{S_1^2S_2^2}$$

$$= S_1S_2 + \frac{9}{(S_1S_2)^2} - 3\left[\frac{1}{S_1} + \frac{1}{S_2}\right]$$

$$= S_1S_2 + \frac{9}{(S_1S_2)^2} - 3\left[\frac{S_2 + S_1}{S_1S_2}\right]$$

$$= -\frac{3}{2} + \frac{9}{\left(-\frac{3}{2}\right)^2} - 3\left[\frac{-3}{-\frac{3}{2}}\right]$$

$$= -\frac{3}{2} + \frac{9}{\frac{9}{4}} - 3\left[-3 \div -\frac{3}{2}\right]$$

$$= -\frac{3}{2} + 9 \div \frac{9}{4} - 3\left[-3 \times \frac{2}{-3}\right]$$

$$= -\frac{3}{2} + 9 \times \frac{4}{9} - 3\,[2]$$

$$= -\frac{3}{2} + 4 - 6$$

$$= -\frac{3}{2} - 2$$

$$P = -\frac{7}{2}$$

Put values of S and P in (1):

$$x^2 - Sx + P = 0$$

$$x^2 - (-19)x + \left(-\frac{7}{2}\right) = 0$$

$$x^2 + 19x - \frac{7}{2} = 0$$

Multiply by 2 on both sides:

$$2x^2 + 2 \times 19x - 2 \times \frac{7}{2} = 0 \times 2$$

$$\boxed{2x^2 + 38x - 7 = 0}$$

Question 7

Find K if S_1 and $S_1 - 5$ are the roots of $x^2 - 3Kx + 5 = 0$.

$$x^2 - 3Kx + 5 = 0$$

$a = 1, \quad b = -3K, \quad c = 5$

$S_1,\ S_1 - 5$ (roots)

Sum of roots:

$$S_1 + S_1 - 5 = -\frac{b}{a}$$

$$2S_1 - 5 = -\frac{(-3K)}{1}$$

$$2S_1 - 5 = 3K$$

$$2S_1 = 3K + 5$$

$$S_1 = \frac{3K + 5}{2} \qquad .....(i)$$

Product of roots:

$$S_1(S_1 - 5) = \frac{c}{a}$$

$$S_1^2 - 5S_1 = \frac{5}{1}$$

$$S_1^2 - 5S_1 = 5 \qquad .....(ii)$$

Put (i) in (ii):

$$\left(\frac{3K + 5}{2}\right)^2 - 5\left(\frac{3K + 5}{2}\right) = 5$$

$$\frac{(3K)^2 + (5)^2 + 2(3K)(5)}{4} - \frac{(15K + 25)}{2} = 5$$

$$\frac{9K^2 + 25 + 30K}{4} - \frac{(15K + 25)}{2} = 5$$

Multiply by 4 on both sides:

$$4 \times \frac{9K^2 + 25 + 30K}{4} - 4 \times \frac{(15K + 25)}{2} = 5 \times 4$$

$$9K^2 + 30K + 25 - 30K - 50 = 20$$

$$9K^2 - 25 - 20 = 0$$

$$9K^2 - 45 = 0$$

$$9K^2 = 45$$

$$K^2 = \frac{45}{9}$$

$$K^2 = 5$$

$$\sqrt{K^2} = \sqrt{5}$$

$$\boxed{K = \pm\sqrt{5}}$$

Question 8

Find K such that 3 is a root of $x^2 + Kx - 21 = 0$.

As given, 3 is the root of $x^2 + Kx - 21 = 0$ .....(1)

So substitute $x = 3$ in equation (1):

$$x^2 + Kx - 21 = 0$$

$$(3)^2 + K(3) - 21 = 0$$

$$9 + 3K - 21 = 0$$

$$3K - 12 = 0$$

$$3K = 12$$

$$K = \frac{12}{3}$$

$$\boxed{K = 4}$$